

ZENITH RAI

Junior Software Developer | Backend | AI/ML | GenAI

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SUMMARY

Junior developer with 3 months of self-directed, project-based experience building applications across machine learning, deep learning, generative AI, and backend engineering. Experienced with FastAPI, PostgreSQL, Docker, and REST API development, with hands-on experience building RAG systems and ML inference applications. Seeking an internship or junior software engineering role where I can contribute to real-world projects and continue developing strong backend and AI engineering skills.

EXPERIENCE

AI / Machine Learning Intern

Sep 2026 – Nov 2026 (3 months)

- Built supervised machine learning models including multilinear regression, logistic regression, decision trees, XGBoost, and AdaBoost using feature engineering and hyperparameter tuning on real-world datasets.
- Developed deep learning models using PyTorch and Keras, including ANN and CNN architectures with Adam and RMSProp optimizers, batch normalization, and dropout, achieving 96.2% accuracy on a retinal image classification task.
- Developed a full-stack RAG chatbot using LangChain, ChromaDB, FastAPI, and HTML/JavaScript, and implemented K-Means and DBSCAN clustering for market segmentation.
- Containerized projects with Docker, managed PostgreSQL database schemas using Alembic, and implemented JWT and bcrypt authentication across backend applications.

PROJECTS (ALL ON GITHUB · ALL DEMONSTRABLE)

Production RAG Chatbot Platform — ZENITH-RAI/rag-chatbot-platform

[FastAPI] [PostgreSQL] [Pydantic v2] [Alembic] [JWT] [bcrypt] [LangChain] [ChromaDB] [Sentence-Transformers] [Llama 3] [Prompt Engineering] [React/HTML+JS] [Docker]

- GenAI** • Built a modular RAG platform with multi-stage document ingestion, dense-vector embeddings, and BM25 hybrid search for retrieval across PDFs and technical documentation.
- Containerized the application using Docker Compose with separate API, cache, and vector storage services and optimized the query-to-generation pipeline.

✓ Live demo available on request

Enterprise E-Commerce REST API Backend — ZENITH-RAI/ecommerce_backendprj

[FastAPI] [PostgreSQL] [SQLAlchemy 2.0] [Alembic] [Pydantic v2] [JWT] [bcrypt] [Docker] [PyTest]

- Backend** • Developed a modular REST API using FastAPI, Pydantic v2, SQLAlchemy 2.0, and PostgreSQL with JWT authentication, bcrypt password hashing, and role-based access control.
- Managed database schema changes with Alembic and wrote PyTest tests covering 90%+ of API endpoints.

✓ Live demo available on request

Deep Learning Retinal Disease Detector — ZENITH-RAI/EYEDISEASEPREDICTION

[PyTorch] [Keras] [CNN] [ANN] [Adam] [RMSProp] [Batch Norm] [Dropout] [OpenCV] [Data Augmentation] [FastAPI] [Docker]

- DL** • Trained CNN and ANN models on 4,000+ fundus images across four classes, achieving 96.2% test accuracy using preprocessing and data augmentation.
- Developed a FastAPI inference API with OpenCV-based image preprocessing for real-time model inference.

✓ Live inference demo available on request

Vehicle Valuation & Marketplace Engine — ZENITH-RAI/final-year-project

[Scikit-learn] [XGBoost] [AdaBoost] [Gradient Descent] [Multilinear Reg] [Logistic Reg] [Decision Trees] [K-Means] [DBSCAN] [PostgreSQL] [FastAPI]

- ML** • Developed a vehicle price prediction system using regression, XGBoost, and AdaBoost models, achieving an R^2 score of 0.87, with K-Means and DBSCAN for market segmentation.
- Built the supporting marketplace API with FastAPI, PostgreSQL, and eSewa payment integration.

✓ Live demo available on request

SKILLS

Backend & APIs: Python, FastAPI, REST APIs, Pydantic v2, SQLAlchemy 2.0, PostgreSQL, Alembic, JWT, bcrypt, Asyncio, OpenAPI / Swagger

Generative AI: RAG, LangChain, ChromaDB, Sentence-Transformers, Ollama, Llama 3, Prompt Engineering, AI Chatbots, Hybrid Search, BM25

Machine Learning: Scikit-learn, Supervised Learning, Multilinear Regression, Logistic Regression, Decision Trees, XGBoost, AdaBoost, K-Means, DBSCAN, Feature Engineering, Hyperparameter Tuning

Deep Learning: PyTorch, Keras, ANN, CNN, Adam, RMSProp, Batch Normalization, Dropout, Data Augmentation

Frontend: HTML, CSS, JavaScript

EDUCATION

BSc. Computer Science & Information Technology (CSIT)

Expected Dec 2026

Tribhuvan University, Kathmandu

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Object-Oriented Programming, Database Management Systems, Advanced Database, Data Warehousing & Data Mining, Artificial Intelligence, Web Technology, Software Engineering, Operating Systems, Computer Networks, Cloud Computing, Network Security, and more.